

DEEP LEARNING FUNDAMENTALS

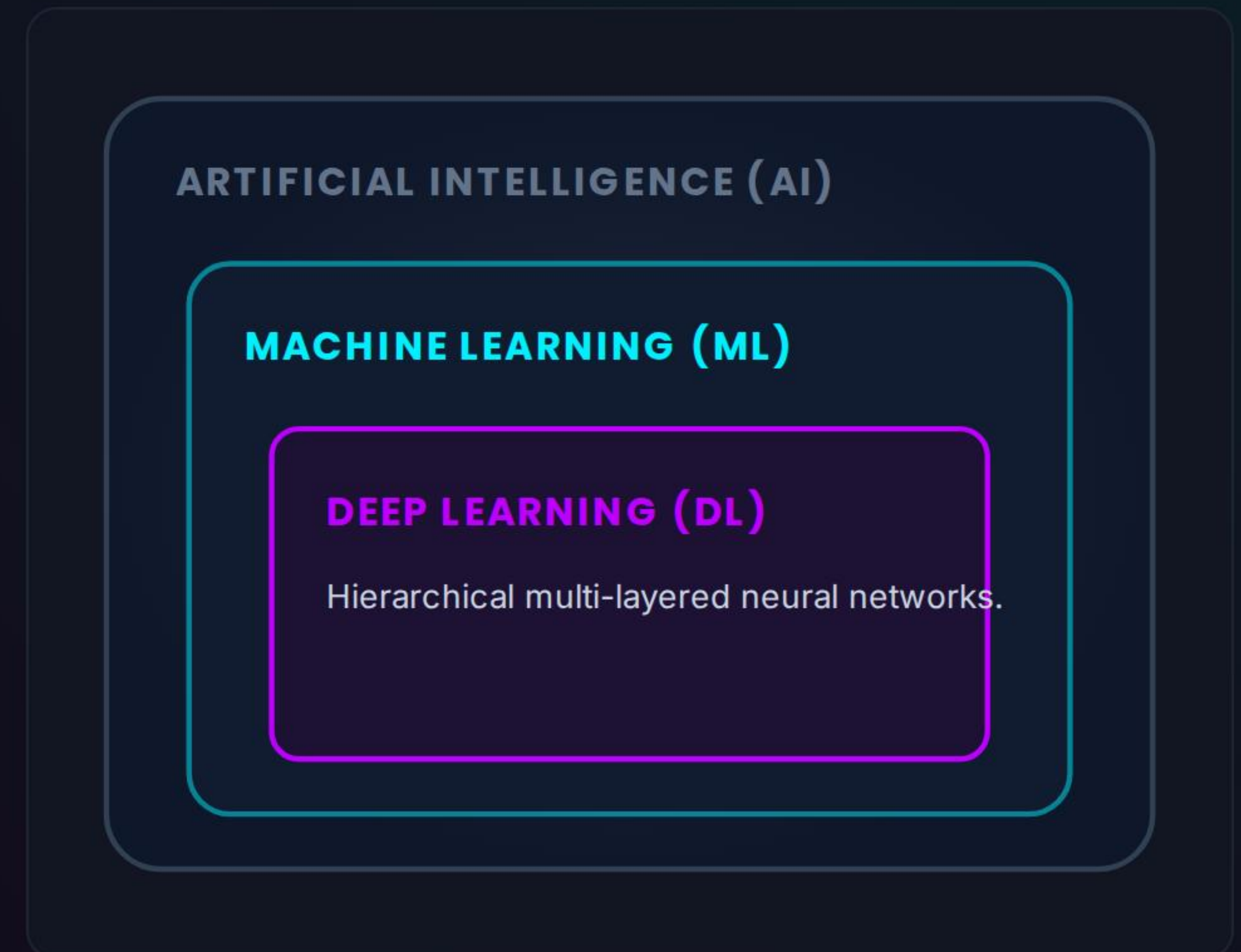
A visual, mathematical introduction to perceptrons, stacked networks, and hierarchical feature representations.

WHAT IS DEEP LEARNING?

Hierarchical Feature Extraction

Deep Learning is a subset of Machine Learning characterized by the use of **Artificial Neural Networks** with multiple hidden layers.

- **Manual vs Automatic:** Classic ML requires human engineering for feature selection. DL learns complex features straight from raw data.
- **Compositional Hierarchy:** Low layers identify basic edges; middle layers combine them into shapes; deep layers recognize highly structured conceptual objects.



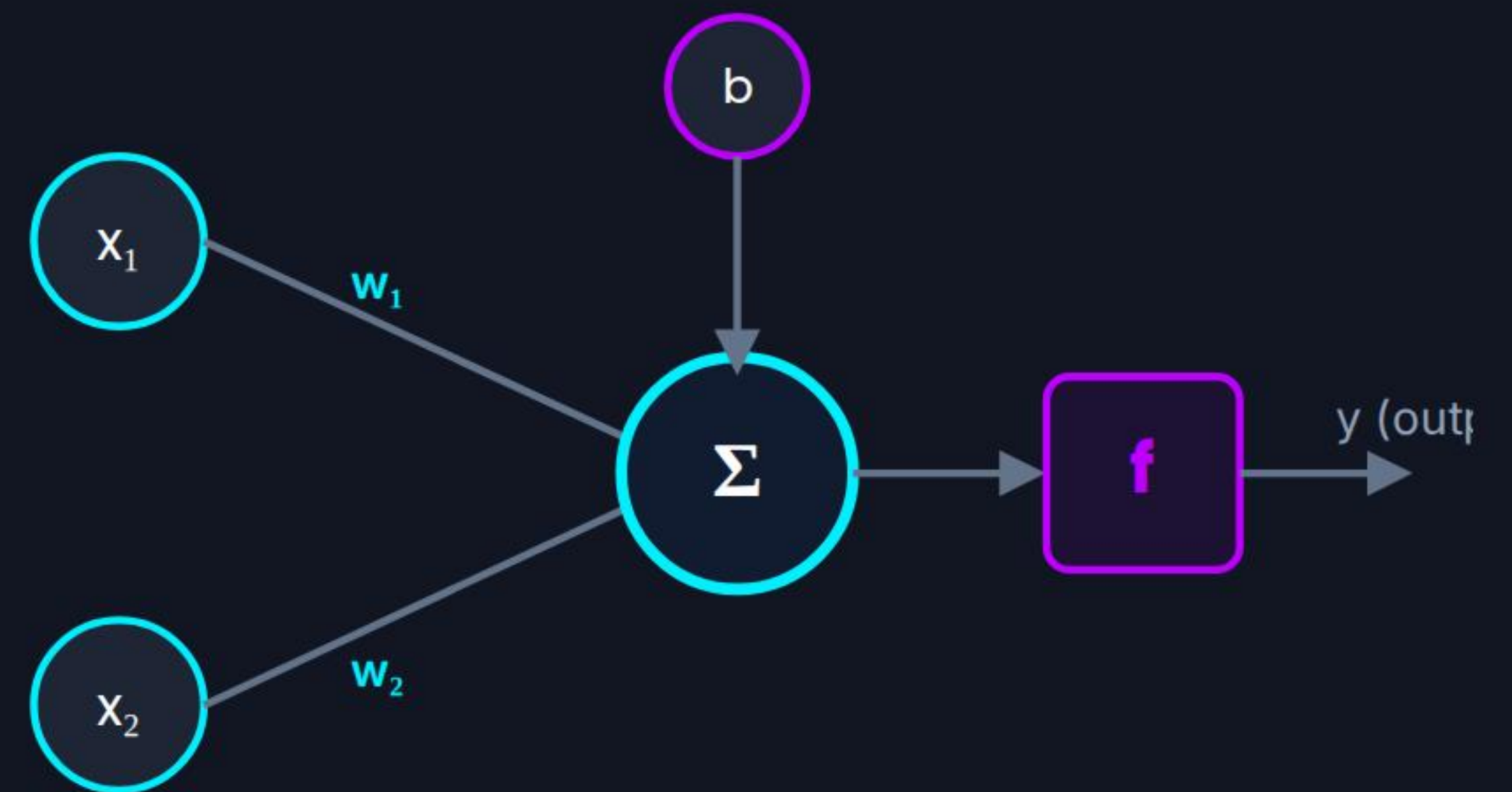
THE PERCEPTRON

The Biological Neuron Analogy

The single perceptron is the elemental building block of all modern deep architectures. It calculates a weighted sum of inputs and applies a threshold.

$$z = w_1 x_1 + w_2 x_2 + b$$

$$a = \sigma(z)$$



ACTIVATION & LOSS FUNCTIONS



Activation Functions

Acts as mathematical gateways to introduce **non-linearity**. Without them, deep networks fold into single linear transformations.



Sigmoid: Maps coordinates cleanly to a (0, 1) probability range.



ReLU: Simple, avoids vanishing gradient issues: $f(x) = \max(0, x)$.



Loss Functions

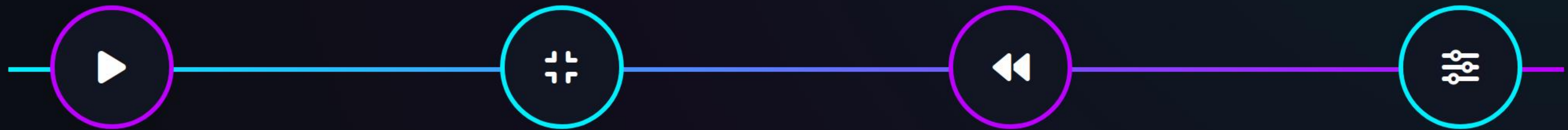
Quantifies the error performance of the model by measuring the exact distance between predictions and actual ground truth.

Binary Cross-Entropy Loss:

$$L = - [y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

By calculating this discrepancy, optimization algorithms obtain the necessary direction to tune model variables.

THE NEURAL TRAINING PIPELINE



1. Forward Prop

Inputs feed forward through hidden layer synapses to generate prediction metrics.

2. Loss Calculation

Loss functions mathematically score model error against the targeted label.

3. Backpropagation

Error derivatives flow backward to compute structural gradients using chain rule.

4. Weight Update

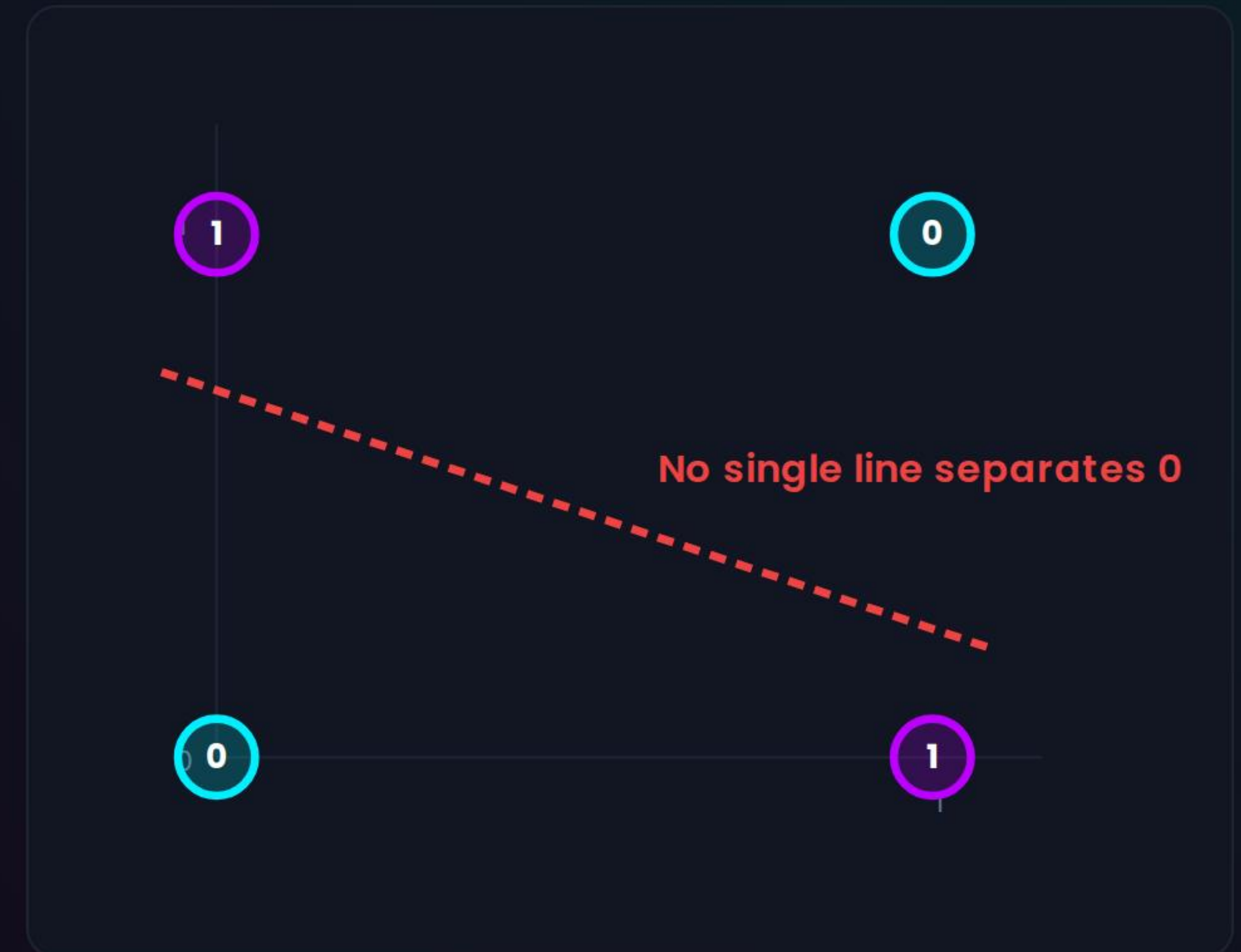
Optimizer (e.g., Adam, SGD) updates connection weights to minimize error paths.

DRAWBACK OF SINGLE PERCEPTRONS

The Linear Separability Limit

Single-layer perceptrons can only classify **linearly separable** patterns. If a straight line cannot divide the classes, the algorithm fails entirely.

- > **The XOR Catastrophe:** In 1969, Minsky & Papert proved a single perceptron could not learn the simple non-linear exclusive-OR logical gate.
- > **The Solution:** By stacking multiple layers of perceptrons, we can model non-linear boundaries.

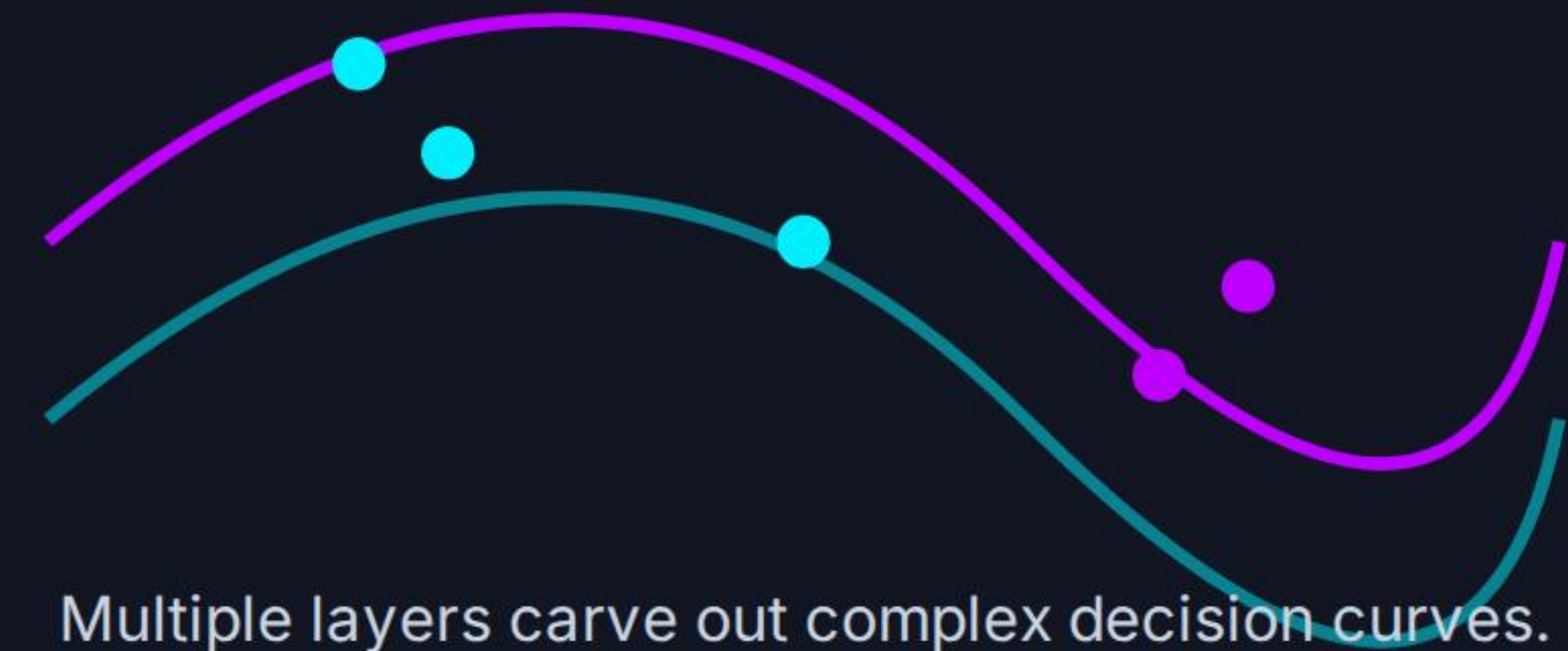


MULTILAYER PERCEPTRONS (MLPS)

Overcoming Dimensional Limits

By cascading neural layers, standard perceptrons transform into highly powerful estimators capable of fitting complex hyper-planes.

- **Universal Approximation Theorem:** A single hidden-layer feedforward network with non-linear activation can approximate any continuous function.
- **Non-Linear Decision Boundaries:** Intermediate hidden nodes warp input feature space so classes become easily separable by output layers.

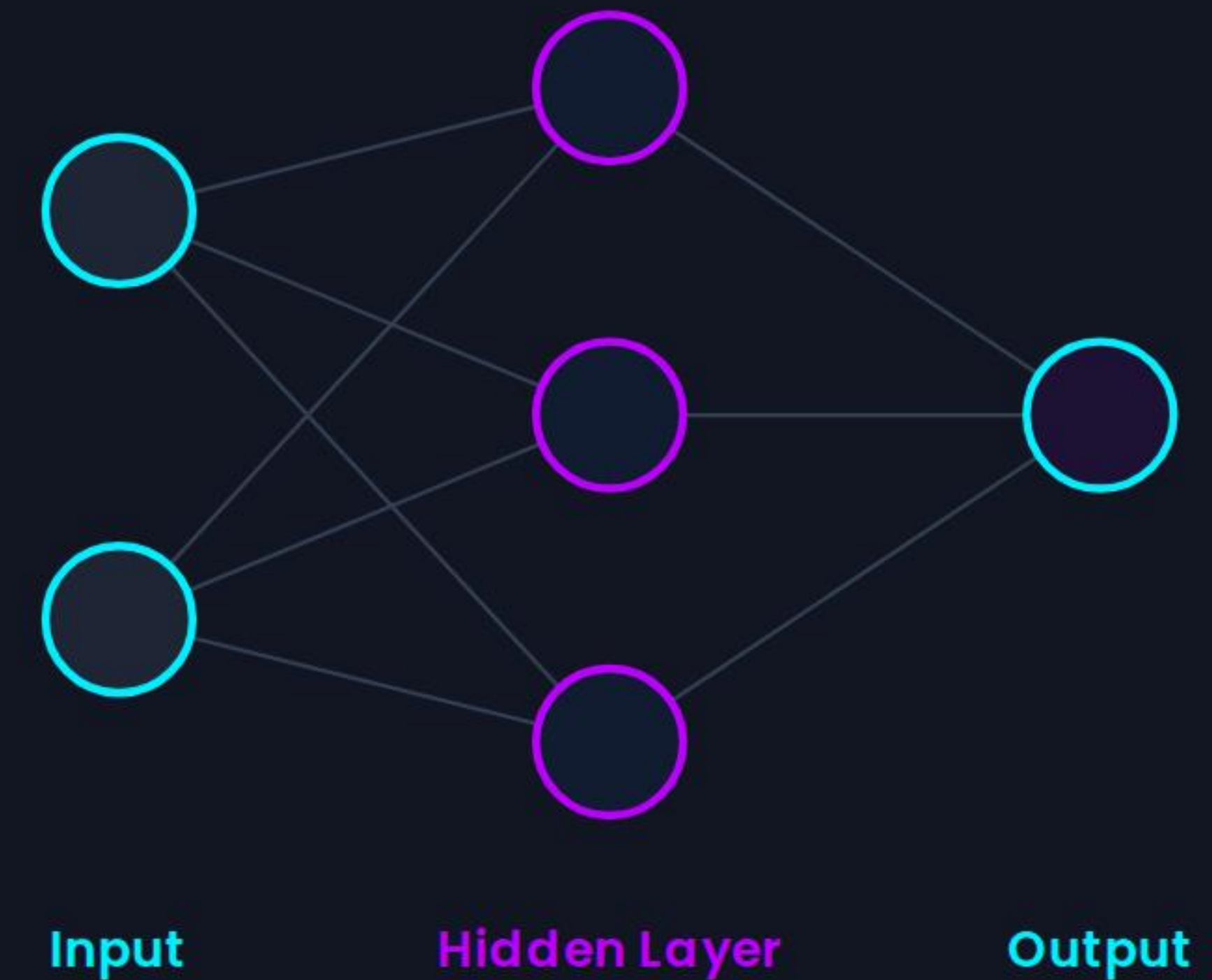


MLP ARCHITECTURE

Fully Connected Feedforward Networks

In an MLP, nodes of a preceding layer connect completely to all neurons in subsequent layers.

- > **Input Layer:** Receives feature vectors directly.
- > **Hidden Layer:** Learns localized abstract features.
- > **Output Layer:** Generates class distributions.

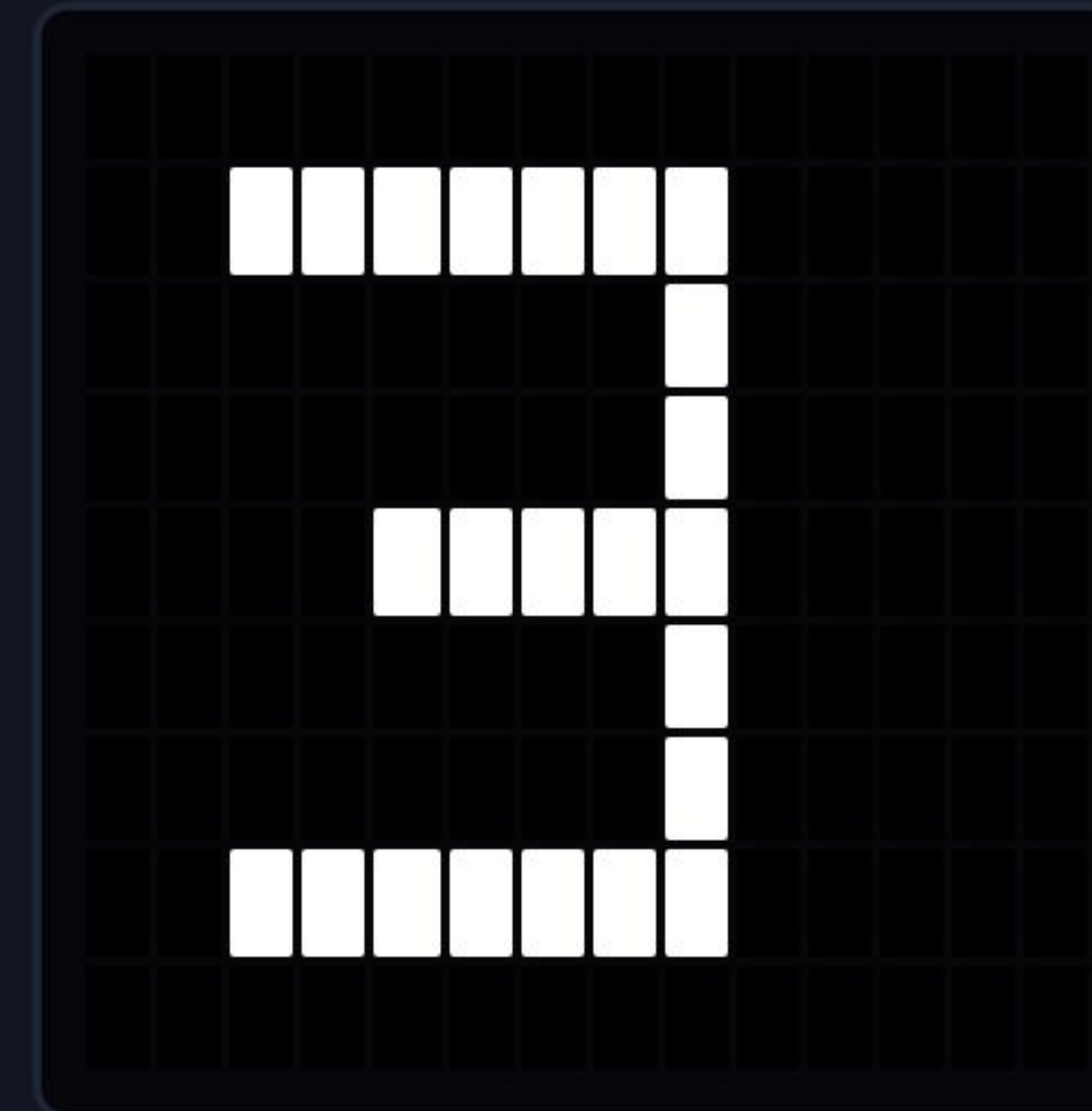


HANDS-ON: MNIST DATASET

The Digits Classification Benchmark

MNIST is the standard introductory computer vision playground. We map grid matrices directly to numerical outputs.

- **Input Specifications:** Handwritten numbers formatted to 28×28 pixel matrices (784 total input nodes).
- **Targets:** Class indices from 0 to 9 representing digits.
- **Processing Strategy:** Flatten matrix pixels into 1D vectors, scale intensity from [0, 255] to [0.0, 1.0].



ACTIVATIONS

```
[0.0, 0.0, 0.0, 0.98,  
...]  
Prediction: Target "3"
```


Summary & Discussion

You are now ready to construct, visualize, and train deep hierarchical networks.

Prerequisites Done:

Perceptrons, Activations, MLP, XOR Solution, MNIST Pipeline.

Next Chapter:

Convolutional Networks (CNNs) for structured spatial data.